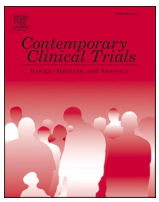
Contemporary Clinical Trials 153 (2025) 107891



Contents lists available at ScienceDirect

Contemporary Clinical Trials

journal homepage: www.elsevier.com/locate/conclintrial





LIVING - protocol for a pragmatic randomized controlled trial investigating the effect of three different levels of diabetes self-management programs

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ARTICLE INFO

Keywords:
Type 2 diabetes
Randomized trial
Diabetes self-management program
High intensity exercise training

ABSTRACT

Background: In Denmark, Type 2 Diabetes (T2D) care includes diabetes self-management programs (DSMPs) delivered by municipalities. Following a successful pilot study, a DSMP was expanded to include a digital platform and flexible educational materials. While exercise therapy is recognized as essential in T2D management, high-quality studies on DSMPs incorporating exercise are limited. This trial assesses the (cost)effectiveness of three interventions with escalating intensity, hypothesizing that higher intensity enhances diabetes management competence and reduces complication risk.

Methods/design: A pragmatic randomized controlled trial will assess three different DSMP interventions in six Danish municipalities acting as trial sites. Each site will implement one of three versions of Lev Livet: a short version (SLL), a full version (FLL), or the full version with high intensity exercise (FLL + H). At each site, 48 participants will be randomly assigned (2:1) to intervention or control groups. Primary outcome is perceived competence in diabetes (questionnaire). Other outcomes measure physical activity level (accelerometer),HbA1c, disease progression, physical function, and self-reported metrics. The trial is approved by the Danish National Center for Ethics (case no 1–10–72-107-23) and registered at ClinicalTrials.gov (NCT06091501).

Discussion: The LIVING trial has the potential to establish whether the Lev Livet program is effective and cost effective and how it can be implemented in municipalities. By analyzing dose-response relationships, the trial may refine intervention frequency and intensity, improve diabetes management, and inform both national and international guidelines for accessible, effective diabetes care.

1. Introduction

1.1. Background and rationale

Lifestyle interventions are a cornerstone of clinical care for individuals with Type 2 Diabetes (T2D). In Denmark, municipalities are currently responsible for providing patient education and diabetes self-management programs (DSMPs) for individuals living with T2D [1]. However, the Danish government's new healthcare reform will transfer this responsibility to the regional authorities in 2026 [2]. This shift represents a significant change in how DSMPs will be organized and delivered, potentially creating opportunities to standardize these interventions across regions, particularly in integrating exercise components. The potential to make DSMPs more uniform nationally is however limited by the lack of evidence supporting the (cost)effectiveness of current DSMPs as they vary in their content and emphasis on physical activity (PA) [3].

In January 2024, the Danish Health Authority published a national quality plan for municipal health efforts, introducing guidelines aimed at ensuring high-quality, geographically uniform preventive measures for chronic diseases, including T2D [4]. However, the limited literature supporting these guidelines may raise concerns about their effectiveness, particularly regarding the integration of exercise into DSMPs. This highlights the need for further research to evaluate the real-world impact of DSMPs, both with and without exercise, to inform future intervention development and implementation.

1.2. Live with Diabetes (Lev Livet)

The Lev Livet concept was introduced in 2019 to standardize DSMPs

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https://doi.org/10.1016/j.cct.2025.107891
Received 11 November 2024; Received in revised form 18 March 2025; Accepted 28 March 2025
Available online 29 March 2025
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for people with T2D. Designed as a framework, Lev Livet focuses on coping skills and illness management [5]. In 2020–2021, it was tested in five municipalities and evaluated by the National Institute of Public Health, demonstrating that a standardized DSMP is feasible and improving consistency across municipalities [6]. The evaluation showed high satisfaction among participants (n = 67), all reporting increased knowledge of T2D and healthy habits. 49 % appreciated the group format as it allowed interaction with others facing similar challenges. The evaluation noted that the Lev Livet program needed adjustments before wider implementation. Notably, future versions of the program should be more flexible in terms of the educational material and methods to improve implementation and transferability. Lev Livet should form the framework for a DSMP with a "catalogue" of relevant material and exercises that individuals can choose from. Hence, we have further developed the Lev Livet program and the intervention is now hosted on a website (www.levlivet.online) designed to meet the challenges identified in the feasibility study. Additionally, due to tight resources in some municipalities, a short version was developed, reducing the sessions from nine to six by consolidating four coping-focused sessions into one (www.levlivet.online). Details on the content of Lev Livet can be found in appendix 1 [7–15].

The evaluation lacked surrogate markers for disease progression, such as glycemic control, blood pressure and PA, leaving its impact on reducing diabetic complications unclear. Additionally, exercise was not consistently included, so the evaluation focused exclusively on educational materials.

1.3. High Intensity Interval Training (HIIT)

While exercise interventions often improve glycemic control in the short term, benefits frequently diminish by six months [16], indicating challenges in sustaining long-term adherence. Identifying optimal exercise types and intensities to reduce disease progression and improve adherence is essential, with time commitment being a major barrier [17]. Time-efficient exercise interventions may enhance adherence [17].

HIIT consists of high-intensity bursts (above 80 % of maximal heart rate) with low-intensity recovery periods. Time-efficient HIIT protocols demonstrate equal or superior benefits in glycemic control, physical fitness, and body composition compared to endurance training for those with T2D [18–20]. Most studies focus on lower-body exercises (e.g., cycling or treadmill), though research suggests that insulin sensitivity increases only in trained muscles, potentially decreasing in others [21,22]. Thus, full-body HIIT may enhance metabolic health effects for T2D. HIIT has proven safe, with adherence rates over 80 % in recent meta-analyses [23].

Exercise therapy within DSMPs shows promise for T2D treatment, though further evidence on intervention dosage and the benefits of exercise therapy is needed. This trial evaluates the effectiveness and cost-effectiveness of three interventions, each with distinct intensities: (1) Short version Lev Livet (SLL), (2) Full version Lev Livet (FLL), and (3) Full version Lev Livet + HIIT (FLL + H). We hypothesize that increasing intervention intensity will progressively improve participants' competence in diabetes management and lower their complication risk.

1.4. Objectives

Primary objective: To investigate the effectiveness of three varying intervention doses on perceived competence in diabetes and PA among individuals with T2D compared to no treatment after three months.

Secondary objective: To investigate the dosing effect of the interventions on markers for disease development and to establish the cost-effectiveness of the interventions.

2. Methods

2.1. Trial design and setting

The trial will be carried out as a pragmatic randomized controlled trail design as this allows evaluation of outcomes for complex healthcare interventions in a "real world" setting [24]. The framework for the trial was inspired by the guidelines for developing and evaluating complex intervention provided by the Medical Research Council Guidance [25] and the current article adheres to the SPIRIT statement [26]. The primary outcome is diabetes competences measured by the five-item Perceived Competence in Diabetes scale (PCD-5) before and after the interventions. The trial sites are to deliver either SLL, FLL or FLL + H as interventions with control groups receiving no treatment. Within each trial site, participants will be randomly allocated (2:1) to intervention or a control group. Regardless of their group allocation, participants will continue to receive pharmacological treatments as prescribed by their GP as well as usual GP visits. The flow of participants is described in Fig. 1.

The RCT is registered at www.clinicaltrials.gov (ID NCT06091501) and has received approval from Danish National Center for Ethics, case no 1–10–72-107-23 before the initiation of any trial procedures. Additionally, all participants will provide written informed consent prior to enrolment, and data processing agreements with the trial sites will be secured in advance.

Six municipal trial sites located in Region Zealand, Central Region and Southern Region of Denmark will provide the intervention and be the place of trial execution and data collection. All data will be analysed by Steno Diabetes Center Aarhus.

2.2. Trial sites

All municipal healthcare centres within the Central Denmark Region were invited to participate in the RCT. In the Northern Denmark Region, Southern Denmark, and Region Zealand, only healthcare centres with a sufficiently large flow of citizens to complete the trial within 18 months were invited. To date, four trial sites have committed to participate in the trial: two sites in the Central Denmark Region, one in Region Zealand, and one in Southern Denmark. Among these, one site offers SLL, two sites offer FLL, and one site provides the FLL + H. This distribution across regions and intervention types ensures a diverse representation of intervention offerings, which will enhance the generalizability of the trial's findings.

2.3. Participants

A total of 288 participants (n = 48 at each trial site) with T2D fulfilling the eligibity criteria listed below will be included in the RCT:

Inclusion criteria:

1. Diagnosed with T2D
2. Age ≥18 years
3. Able to speak and understand Danish
4. Individuals assessed fit by a health care professional to attend a DSMP

Exclusion criteria:

Contraindication to increasing level of PA assessed by the participant' general practitioner (GP), e.g. unstable angina, uncontrolled arrhythmias, or severe aortic stenosis.

2.4. Recruitment procedure

The trial flow is shown in Fig. 1. Participants will be recruited through trial sites, with eligibility assessed during a clarifying interview. Citizens may be referred to municipal health services by hospitals, GPs,

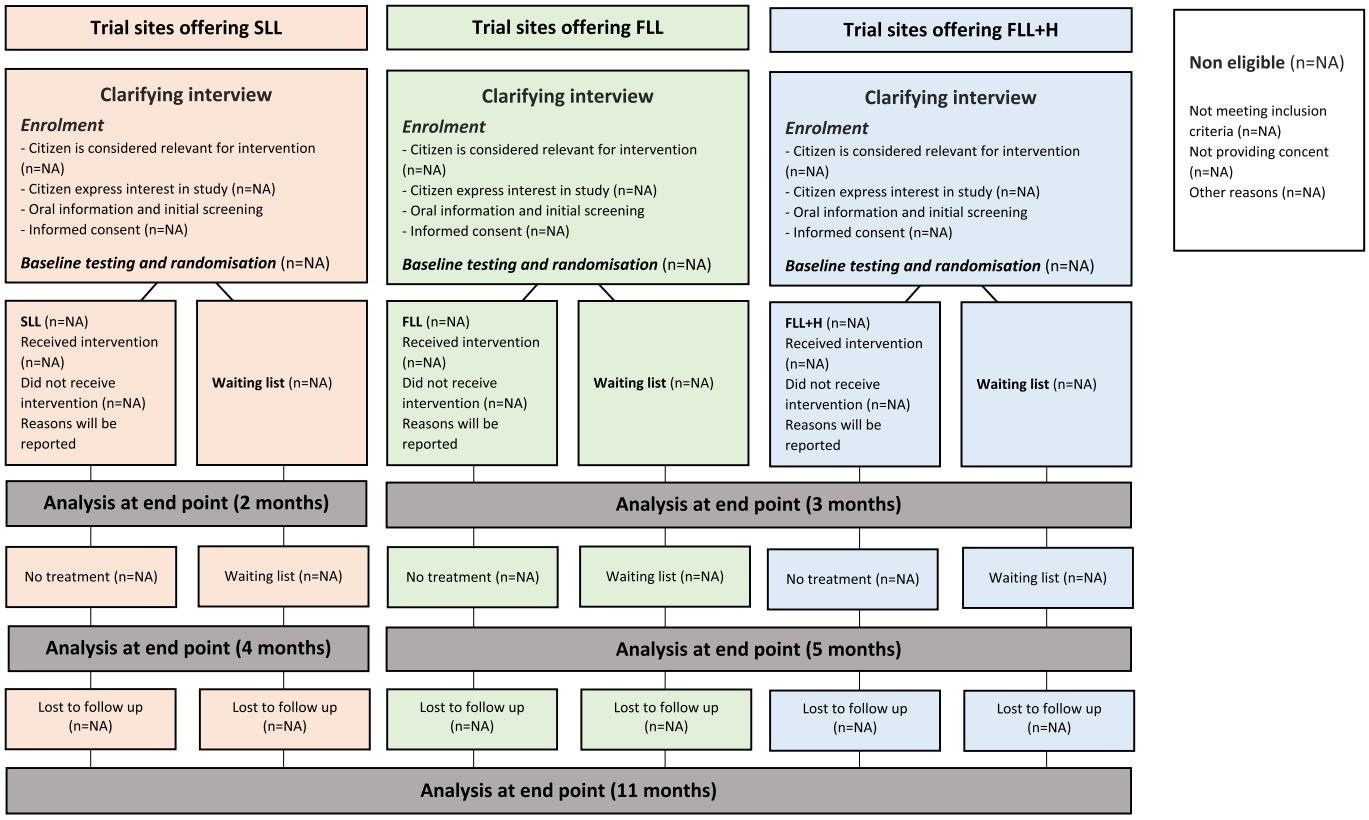


Fig. 1. Trial flow chart.

or self-referral, with most referrals coming from GPs. If participation in Lev Livet is deemed appropriate for the citizen, (s)he is informed about the LIVING research project. Health professionals at the trial sites assess eligibility and are instructed to invite all eligible individuals to participate. Those who decline participation can still participate in Lev Livet but do not get insights into relevant test results. During the interview, health professionals provide verbal and written information about the project, including the citizen's right to a 10-day deliberation period. Informed consent is obtained either during the interview or by the first Lev Livet session.

2.5. Randomization

Randomization will be done at the individual level. The trial sites will join the intervention that is best suitable to their context and resources.

Following informed consent, the participants will be randomly allocated (2:1 allocation ratio) to intervention or control group using the Research Electronic Data Capture (REDCap) system. A priori, an independent statistician has prepared a computer-generated randomization schedule in randomized blocks of 12 persons.

2.6. Blinding

Due to the nature of this real-life pragmatic trial, neither the participant nor the health professional responsible for data collection can be blinded after the allocation.

2.7. Interventions

2.7.1. Full Lev Livet (FLL)

Interventions are shown in Fig. 2. Participants allocated to FLL will participate in a total of 9 sessions with a duration of two and a half to three hours for each session. In the first four sessions, coping skills are taught across different chronic diseases. In this part of the intervention citizens suffering from different chronic diseases (Chronic obstructive pulmonary disease, heart disease and T2D) can participate. Citizens living with T2D then move on to four diabetes-specific sessions, followed by a ninth follow up session. The sessions are intended to be held consecutively over eight weeks, with the ninth session scheduled three weeks after the eighth. However, some flexibility is allowed, as holidays and vacations at the trial sites may slightly affect the timeline.

2.7.2. Short version Lev Livet (SLL)

Participants allocated to SLL will receive a condensed version of Lev Livet. In this version, selected themes from modules 1–4 are provided in a single session, focusing on essential knowledge and tools that support participants in managing life with a chronic illness.

2.7.3. Full Lev Livet + HIIT (FLL + H)

Participants allocated to FLL + H will receive FLL alongside 16 one-hour sessions of supervised exercises with a focus on HIIT. The HIIT framework provides flexibility for physiotherapists to design exercises that meet individual participants' skills, needs, and motivations. Physiotherapists can adjust exercises as long as the prescribed intensity, training volume, and both upper and lower body engagement are maintained.

The 8-week supervised HIIT protocol consists of 5 × 1-min high-intensity intervals, each followed by 1 min of active recovery (Fig. 3). Between blocks of intervals, there will be a 4-min break to allow participants to switch between upper and lower body exercises. The intervention progresses from two training blocks (22 min) in the first two weeks to four training blocks (48 min.) in the final four weeks. Training intensity will be monitored using the Borg 15 scale, aiming for an intensity of Borg 14–16, corresponding to 77–93 % of maximal heart rate. As the primary focus is on HIIT, HIIT will occupy a greater portion of the

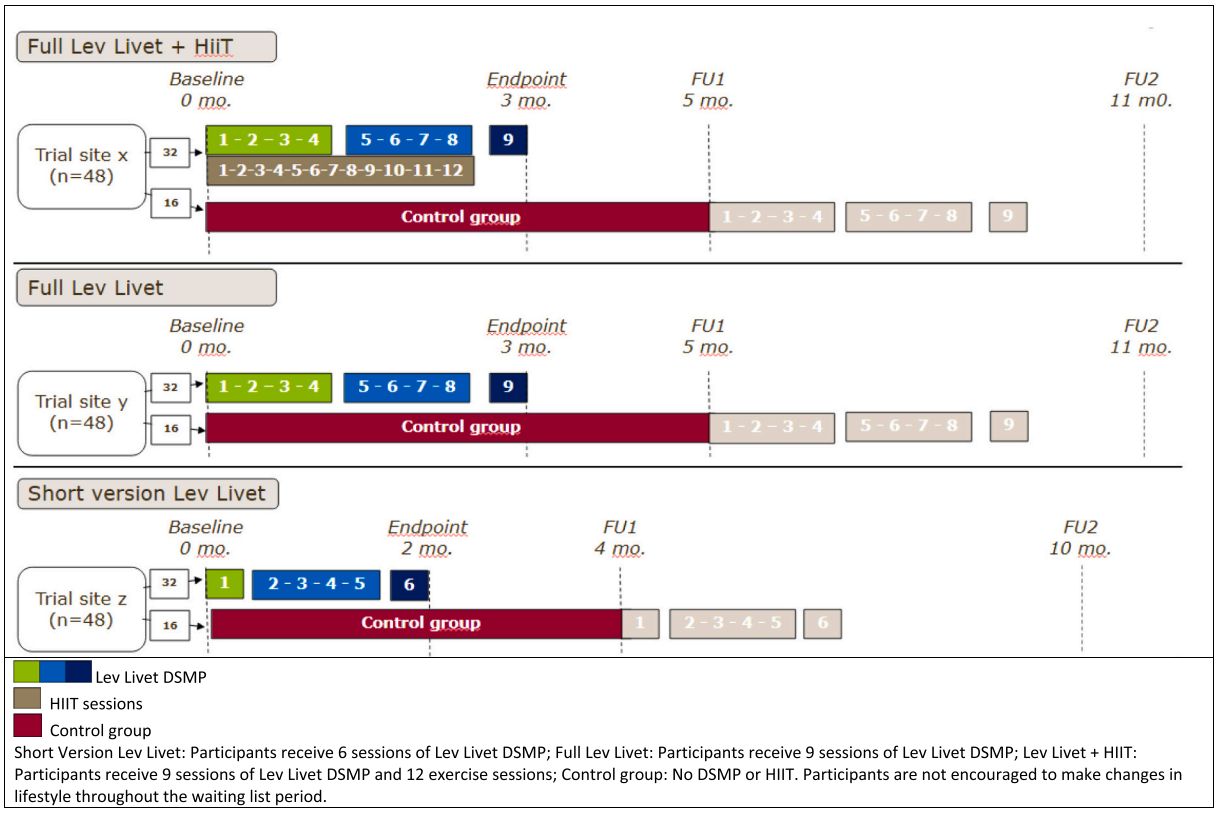
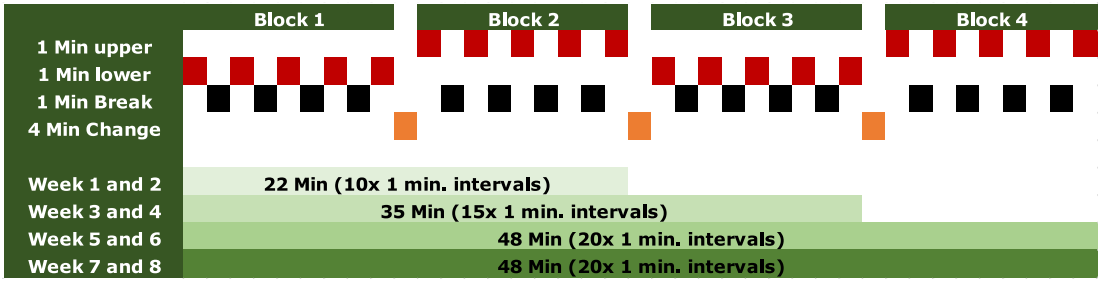


Fig. 2. Flow of participants in the LIVING trial. SLL, Short version Lev Livet; FLL, Full Lev Livet intervention; LL + H, Lev Livet intervention + HIIT training. The clarifying interview is the initial contact with potential participants. Eligability is based on a health professional's assesment on the citizens ability to participate in a group based intervention. If inclusion criteria are met and informed consent is obtained the participant proceeds to baseline testing and randomization. Participants are randomly allocated to intervention or control group in a ratio of 2:1.



Overview of the 8-week supervised HIIT protocol. Each session consists of 5 x 1-minute high-intensity intervals, followed by 1 minute of active recovery. A 4-minute break separates blocks, allowing participants to alternate between upper and lower body exercises. The program progresses from two blocks per session in the first two weeks to four blocks in the final four weeks.

Fig. 3. Framework for HIIT protocol.
Overview of the 8-week supervised HIIT protocol. Each session consists of 5 × 1-min high-intensity intervals, followed by 1 min of active recovery. A 4-min break separates blocks, allowing participants to alternate between upper and lower body exercises. The program progresses from two blocks per session in the first two weeks to four blocks in the final four weeks.

one-hour sessions as the intervention progresses. The remaining time will be spent on exercises aimed at injury prevention, such as balance and strengthening exercises, with a familiarization period for the exercises prioritized during the initial weeks of the intervention. Each session must include:

- Warm-up (5 min).
- HIIT (progressively increasing).
- Cool-down (5 min).

In addition to the two supervised training sessions, the participants will be instructed to independently complete a training session equivalent to the supervised training session for that week.

2.8. Control Group

Participants allocated to the control group are placed on a waiting list. The duration of the waiting list will be aligned with the length of the intervention and is determined by the timing of the first follow-up after

the intervention ends — four months for SLL and five months for FLL and FLL + H. At the clarifying interview, the control group will not receive counselling. The clarifying Interview will focus exclusively on enrolment, randomization and testing.

2.9. Retention

Each trial site will appoint a contact person who will maintain close communication with our team to address any issues that may arise during the project period, such as randomization, data collection, and protocol adherence. To ensure high response rates, both questionnaire and biometric data collection at baseline, endpoint, and first follow-up (FU1) will be managed by the staff at each trial site. For participants unfamiliar with electronic devices, staff will assist with completion of the questionnaire.

2.10. Trial endpoint

An overview of the outcome assessment is found in Table 1.

2.11. Primary outcome

The between-group differences for change in the five item Perceived Competence in Diabetes (PCD-5) from baseline to endpoint.

2.12. Secondary outcomes

The main secondary outcomes are changes in Hba1c and accelerometer derived PA measures (time/day in moderate-to-vigorous physical activity (MVPA), time/day spend being physically active with at least light intensity, number of steps per day and sleep).

2.13. Other exploratory outcomes

Other outcomes designated to investigate effects on self-reported measures (eg. physical and mental health, sleep quality, smoking etc.), biometric outcomes (eg. blood pressure and body composition), and physical function.

2.14. Data collection

The trial schedule (Table 1) follows SPIRIT guidelines. Data are collected at baseline (0 months), endpoint (3 months), FU1 (5 months), and second follow-up (FU2) (11 months) for FLL/FLL + H groups and controls, and at baseline, endpoint (2 months), FU1 (4 months), and FU2 (11 months) for SLL and controls. Testing occurs at each trial site by trained personnel (for a description of the standard procedures, see jonasmunkjensen.github.io/LIVING/). Staff are trained in fingerstick testing, accelerometer fitting, blood pressure measurement, and testing of functional ability. Self-reported data, including the primary outcome, are collected via REDCap, with participants receiving links via E-boks.com. Health professionals send questionnaires at baseline, endpoint, and FU1, while SDCA handles FU2. Staff may assist participants if questionnaires are not completed on time.

2.15. Primary outcome

2.15.1. Perceived Competance in Diabetes (PCD-5)

PCD-5 is a five 5-item scale with seven response categories from not at all true to very true, assessing the degree to which the individuals feel they can manage their diabetes [27,28].

2.16. Main secondary outcomes

2.16.1. Physical activity

PA will be objectively measured at both baseline and endpoint using

Table 1
Outcome measures during the intervention (SPIRIT figure).

	Time points	Baseline	Endpoint	FU 1	FU 2
		0	2 and 3*	4 and 5*	11
Intervention period		X	X		
Participant characteristics					
Demographics		X			
Medicine		X	X	X	X
Comorbidity		X			
Primary outcomes					
Perceived competence in diabetes management	PCD-5	X	X	X	X
Physical activity behaviour Min./day of at least light PA	Accelerometer	X	X		
Secondary outcomes					
Physical activity behaviour Moderate to vigorous PA Vigorous PA Moderate PA Light PA Sedentary activity Steps per day	Accelerometer	X	X		
Glycated haemoglobin	Fingerstick test	X	X	X	
Other outcomes					
Self-reported outcomes					
Quality of life (QOL)	SF1 questionnaire	X	X	X	X
General health & QOL	EQ-5D-5L	X	X	X	X
Wellbeing	WHO-5 Questionnaire	X	X	X	X
Emotional distress	PAID-20	X	X	X	X
Patient engagement	PAM-13	X	X	X	X
Sleep		X	X	X	X
Smoking		X	X	X	X
Loneliness		X	X	X	X
Physical activity in daily life	IPAQ	X	X	X	X
Motivation to exercise	BREQ-2	X	X	X	X
Screening for complications	Frequency	X	X	X	X
Objectively measured outcomes					
Blood pressure	Resting systolic and diastolic blood pressure	X	X	X	
Body composition					
Bioimpedance	Weight, BMI	X	X	X	
Waist/hip	WH	X	X	X	
Physical function					
Leg strength and endurance	30-CST	X	X	X	
Aerobic capacity and endurance	6mw	X	X	X	
Sleep	Accelerometer	X	X		

* In SLL, endpoint and FU1 occur one month earlier than in FLL and FLL + H

Abbreviations: FU (follow up), PCD-5 (Perceived Competence Scale for Diabetes), PA (physical activity), SF-1 (Short Form 1 item questionnaire), EQ-5D-5L (EuroQol – 5 level version), WHO-5 (The World Health Organization – five well being index), PAID (Problem Areas In Diabetes), PAM-13 (Patient Activation Measure), IPAQ (International Physical Activity Questionnaires), BREQ-2 (The Behavioural Regulation in Exercise Questionnaire), BMI (body

mass index), WH (Waist/hip circumference), 30-CST (30 s Chair Stand Test), 6mw (6 min walk test).

accelerometers (AX3, Newcastle, UK) worn on the right thigh. Participants will be instructed to wear the accelerometers for seven consecutive days, 24 h per day, including during sleep and any land based or water activities. The following aspects of PA will be assessed

- Time/day spent in MVPA
- Time/day spent being physically active with at least light intensity
- Time/day in sedentary activity
- Number of steps per day

2.16.2. Hba1c

Blood sampling will be conducted at all trial sites at baseline, endpoint and FU1 using a closed system called Cobas B 101 and standard procedures. A trial nurse will carry out the fingerstick test. A single drop of blood (1.5 μl) will be collected and immediately analysed for HbA1c and registered in REDcap.

2.17. Other outcomes

2.17.1. Objective outcomes

Objective outcomes will be collected at baseline, endpoint and FU1. For a detailed description of the standard operation procedure, see jonasmunkjensen.github.io/LIVING/2-testmanual.

Blood pressure will be monitored by standard procedure for home blood pressure measurements using the machine used by the health care centre. Systolic and diastolic blood pressures after 5 min of rest will be measured as the lowest of two measurements.

Weight, body fat and BMI are measured using bioelectrical impedance Tanita body composition analyses.

Waist/hip circumference and height will be measures using standard procedures.

Six-minute walk is a standardized, self-paced walking test used to asses daily physical capacity in individuals with T2D [29]. The test will be performed on a flat floor over a 25-m track. Participants receive verbal instructions to walk the track as fast as possible without running for six minutes. The distance walked in meters will be recorded.

The Chair stand test (30-CST) is an uncomplicated test and widely used to measure strength and endurance in lower limbs and providing information about fatique [30]. Participants receive instructions to get up and sit down from a chair as many times as possible in 30 s.

2.17.2. Self-reported measures

Self-reported measurement will be collected at baseline, endpoint, FU1 and FU2.

We will assess quality of life and general health with the SF-1 and EQ-5D-5L [31]. Mental well-being will be measured using the WHO-5 Well-Being Index, which evaluates mood, vitality, and interest in life [32]. The PAM-13 will gauge participants' knowledge, skills, and confidence in self-management [33], while diabetes-related psychosocial challenges will be captured with the PAID-20 scale [34,35].

PA levels over the past week will be recorded using the IPAQ, classifying activity as low, moderate, or high [36]. To assess motivation for PA, we will use the BREQ-2, grounded in self-determination theory [37,38].

Demographic and health data—including education, civil status, occupation, diabetes diagnosis time, medication, and screening history for diabetes complications—will be collected alongside information on smoking, sleep patterns, and loneliness. These factors will serve as covariates in the analysis to account for influences on health outcomes.

2.17.3. Health economical evaluation

We will conduct a cost-effectiveness analysis of the three interventions, relative to the incremental effect on the primary outcome. Costs will be estimated by identifying relevant components, measuring resource use, and applying unit prices. The incremental cost-effectiveness ratio (ICER) will reflect the cost per point gained on the PCD-5 scale. Deterministic and probabilistic sensitivity analyses will assess the ICER's reliability. In the cost effectiveness analysis we take the perspective of the municipalities as the expected economic consequences in the rest of the health system and society as a whole is expected to be negligeable.

2.17.4. Process evaluation

We will conduct a process evaluation of the health care professsionals' perspectives focusing on implementation fidelity according to Steckler and Linnan (2002) [39]. Research questions on the degree to which healthcare professionals appraise the intervention and react to the activities included will guide data collection. Further, the municipalities register number of participants invited and included and attendance.

2.18. Power calculation

To estimate site-specific effects, we calculated power based on an assumed between-group difference of 5 points on the PCD-5 scale (range: 5–35) with an individual change standard deviation (SD) of 4. This assumption is supported by findings from Varming et al. [27], whose intervention—similar to ours—led to a within-group increase of 5 points in PCD-5, which we consider a clinically meaningful difference. A 2:1 allocation ratio (intervention: control) was specified. Incorporating a design effect of 1.7 to allow for clustering effects due to intervention groups, we determined that each municipality should randomize 48 participants (32 intervention, 16 control) to achieve 80 % power. The design effect was based on an average number of 12 individuals in each group and an assumed ICC of 1.7 [40].

2.19. Statistical analysis methods

2.19.1. Analysis of primary outcomes

The primary analysis will focus on the mean difference in individual changes from baseline to endpoint, following the intention-to-treat principle. Accordingly, all participants randomized to an active treatment group will be analysed as part of that group, regardless of compliance. Missing data in the primary analysis will be managed using multiple imputation. Adherence will be deemed satisfactory if participants attend $\geq$7 of 9 Lev Livet sessions (FLL), $\geq$12 of 16 supervised sessions (FLL + H), and $\geq$ 5 of 6 sessions (SLL).

Primary outcomes will be assessed using a mixed model with repeated measures. Time will be included as a categorical covariate, alongside randomization status and their interaction, enabling comparison of outcomes at each follow-up time point across groups, as well as an evaluation of overall differences. Dependence of repeated measures within participants will be addressed by including a random effect for participants. Additionally, random effects for group and trial site will be incorporated, assuming independence between random effects at each level (participant, group, trial site). Baseline equivalence across intervention and control groups will be assessed including covariates in the analyses to increase outcome precision if relevant. In a supplemental analysis, a similar analysis will be performed in individuals who at baseline had a low PCD-5 score (PCD-5 < 30).

A hierarchical analytic approach will guide the analysis with subsequent steps interpreted based on p-values and confidence intervals if initial tests do not indicate significant differences ($p > 0.05$).

1. Individual intervention testing: We will do between group testing between the interventions and their respective control group.

- If any of the comparisons yield significant findings, we will pool the interventions and compare them and the control group.

2. Trend analysis: If between-group differences are identified, we will conduct a trend analysis, investigating differences between the escalating intensities.
3. Follow-up comparison: If a trend is present at the endpoint, we will test differences in the two primary outcomes between the endpoint and FU1.

2.19.2. Analysis of secondary and exploratory outcomes

The secondary outcomes, HbA1c and PA, will be analysed following the same approach as outlined for the primary outcome. Additional exploratory outcomes, including questionnaire and biometric data, will be analysed using mixed models with repeated measures or multilevel logistic regression with random effects, depending on the nature of the outcome. Analyses will be adjusted for relevant covariates, with both within-group and between-group comparisons conducted. Effects will be reported descriptively without formal hypothesis testing.

2.20. Data managing

The REDCap system will be used for data entry in the electronic Case Report Form (e-CRF) and questionnaires. All personal data will be pseudonymized. All information related to participants will be managed in compliance with Danish regulations for personal data protection and Danish health law.

2.21. Monitoring

The project is being initiated and managed in partnership between Steno Diabetes Centre Aarhus (SDCA) and Steno Diabetes Centre Odense (SDCO). SDCA will oversee the final dataset and maintain access to it. A research group has been established to monitor the research process and ensure adherence to the research protocol. The group includes chief clinical researchers, a trial manager and a consultant. The data management team, comprising members of the research group and a statistician, will bring together clinical, research, and statistical expertise. Since the intervention is not anticipated to cause harm or adverse effects, no interim analyses or audits are planned.

2.22. Dissemination plan

Trial results will be broadly disseminated, regardless of whether they favour the interventions, the control groups, or are inconclusive. Our dissemination plan targets key stakeholders through various channels. Regular newsletters will update citizens and health workers at trial sites with project status and preliminary findings. Additionally, periodicals will be sent to municipal decision-makers and general practice staff to ensure local healthcare administrators are informed.

Findings will be presented at relevant national and international conferences to engage the wider research and clinical community. We plan to publish at least three peer-reviewed articles in international journals, focusing on primary and secondary outcomes, following CONSORT guidelines for pragmatic trials. Results will also be shared via online support tools, multimedia, print educational resources, and oral presentations.

Authorship will adhere to ICMJE criteria. Significant protocol amendments will be reported to the Central Region's Health Research Ethics Committee and updated on ClinicalTrials.gov.

3. Discussion

This project evaluates the (cost)effectiveness of three versions of Lev Livet to advance diabetes self-management and exercise therapy. Previous assessments found Lev Livet feasible and well-received, laying a foundation for this RCT to test its effectiveness in real-world settings [6]. We will assess DSMP combined with exercise, an approach with limited high-quality evidence despite substantial support for exercise as a safe and effective treatment for T2D [17–20,23,41–43]. Adapted from a previous HIIT protocol that demonstrated significant improvements in insulin sensitivity, VO2-max, and body composition with high adherence and no injuries, our HIIT model has been tailored for municipal healthcare centers [44].

This trial addresses a research gap by examining how three different DSMP interventions affects outcomes. This insight will help refine the intervention for optimal benefits and cost-effectiveness, guiding best practices in diabetes care. Gaining a nuanced understanding of the dose-response relationship may enhance personalized diabetes care by identifying which intensity of self-management support and exercise are most suitable for different individuals.

A health economic evaluation will explore resource-efficient models, as many healthcare centers cannot offer the long version of Lev Livet. The condensed short version increases accessibility by maximizing resources, and this trial will determine if the shorter intervention matches the outcomes of the full version.

The trial has some important limitations as well. Health professionals and participants cannot be blinded to the allocation, which may bias the outcomes. Further, the participants allocated to the control group may decide to improve diabetes management or increase their physical activity through different methods and thereby dilute the effects between intervention and control group groups.

The waiting list design makes us unable to evaluate effects beyond five months. The waiting list design was chosen as it would be unethical to prevent the control group from receiving a diabetes management course.

The trial may be prone to selection bias. We risk that only resourceful municipalities with a special interest in T2D participate. Further, evaluating each intervention across three regions may introduce variability due to differences in regional populations and the implementation of the interventions. These factors could limit the comparability of results across the three interventions.

In the short term, we expect improvements in self-management, PA, and HbA1c reduction over 5 mmol/mol. If successful, Lev Livet could become a key component of Denmark's healthcare system and potentially inform international guidelines for accessible, effective diabetes care.

Competing interest and funding

The authors are employed by the SDCA with no additional conflicts of interest to declare. Support for this trial is provided by the SDCA, partially funded through an unrestricted donation from the Novo Nordisk Foundation. Running costs, amounting to approximately 3.2 million DKK, will be covered by SDCA, with additional funding sought for the support of at least one PhD fellow.

CRediT authorship contribution statement

Jonas Munk Jensen: Writing – original draft, Validation, Software, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Karina Vejrum Sørensen:** Conceptualization, Methodology, Writing – review & editing. **Henrik Støvring:** Methodology, Validation, Writing – review & editing. **Anette Andersen:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Trial status

Currently, four trial sites have been recruited and three are collecting data. The last trial site is scheduled to begin data collection in April 2025. The anticipated trial completion date is September 2026. Protocol version: 2, dated 18.03.2025.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

We extend our gratitude to the healthcare centers for helping design a feasible, practical trial and providing valuable feedback on methodologies and outcomes. Thanks also to the trial sites, healthcare providers, and everyone involved in the trial.

Appreciation goes to our steering committee, Professor Annelli Sandbæk and Manager Arne Gaarn; research group members, Professor Søren Skou and Ph.D. Mette Juel Rothmann; and project team members, Vibeke Brinkmann Løjte, and Nicolaj Holm Ravn Faber.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cct.2025.107891.

Data availability

No data was used for the research described in the article.

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